# Canvas Helper AI Assistant
# Requirements Traceability Matrix

Team: Team 16 · Course: COSC 499 · Date: August 2026
Final Delivery — Requirements Implementation & Validation Traceability

## 1. Purpose

This matrix demonstrates that every documented User Requirement (UR) and Functional Requirement (FR) for the Canvas Helper AI Assistant has been implemented and validated. It consolidates milestone coverage (Milestones 1–6), Testing Event 2 PASS results, and final automated/manual testing evidence into a single submission artifact.

Primary validation evidence: Testing Event 2 requirements matrix (all requirements PASS) and the Final Testing Report (Pytest / Vitest / Playwright / Hypothesis). Implementation references point to the delivered application on the final codebase.

## 2. Summary

| Requirement set | Count | Result |
|---|---|---|
| Administrator user requirements (UR1.1–UR1.7) | 7 | **7 PASS / 0 FAIL** |
| Instructor user requirements (UR2.1–UR2.29) | 29 | **29 PASS / 0 FAIL** |
| Functional requirements (FR1–FR13 groups) | 72* | **All PASS** |
| Overall user requirements | 36 | **36 PASS / 0 FAIL** |

*FR line items counted from FR1.1–FR13.5 as recorded in the TE2 matrix (FR8 presented as five consolidated history/reversal items in this document).

## 3. User Requirements Traceability

User requirements are traced to product surfaces and confirmed PASS in Testing Event 2. UR1 covers administrator capabilities; UR2 covers instructor capabilities.

### *3.1 Administrator requirements (UR1.1–UR1.7)*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| UR1.1 | Secure administrator login and access to administrative functions | Admin auth UI/API; JWT sessions; RBAC | TE2; auth/admin tests; e2e admin | **PASS** |
| UR1.2 | Create, update, enable, disable, and manage instructor accounts | Admin instructor management pages/services | TE2; admin management tests | **PASS** |
| UR1.3 | Revoke instructor access when required | Revoke Canvas access / disable account flows | TE2; revoke access tests | **PASS** |
| UR1.4 | Monitor system health and operational status | System Health dashboard + health services | TE2; health/troubleshoot tests | **PASS** |
| UR1.5 | Review audit logs and security-related events | Audit log viewer + audit logging | TE2; audit tests | **PASS** |
| UR1.6 | Review failed Canvas and AI operations | Failed Operations page + failed_operations | TE2; failed-ops tests | **PASS** |
| UR1.7 | Troubleshoot without accessing sensitive credentials | Troubleshooting diagnostics with credential masking | TE2; troubleshoot tests | **PASS** |

### *3.2 Instructor requirements (UR2.1–UR2.29)*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| UR2.1 | Secure account creation, authentication, and application access | Register/login/JWT/protected routes | TE2; auth tests; e2e | **PASS** |
| UR2.2 | Securely connect a Canvas API token | Connect Canvas UI + encrypted token storage | TE2; canvas connection tests | **PASS** |
| UR2.3 | Replace or remove a stored Canvas API token | Settings/connect replace & remove flows | TE2; canvas token tests | **PASS** |
| UR2.4 | View Canvas courses for the connected token | Courses list + Canvas sync | TE2; canvas/course tests | **PASS** |

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| UR2.5 | Select an active course | Active course preference + sidebar selector | TE2; active-course tests | **PASS** |
| UR2.6 | View course info, students, assignments, and submissions | Course tabs + Canvas retrieval services | TE2; canvas data tests | **PASS** |
| UR2.7 | Analyze missing submissions | Missing submissions analysis UI/service | TE2; missing-submission tests | **PASS** |
| UR2.8 | Analyze late submissions | Late submissions analysis UI/service | TE2; late-submission tests | **PASS** |
| UR2.9 | Search and filter course information | Tables search/filter/pagination | TE2; UI/system tests | **PASS** |
| UR2.10 | Access a dashboard summarizing course activity | Instructor dashboard overview | TE2; dashboard tests | **PASS** |
| UR2.11 | Interact using natural-language requests | AI Assistant chat + LLM services | TE2; chat/assistant tests | **PASS** |
| UR2.12 | Ask questions about course data | Read-only assistant/Canvas context tools | TE2; chat/read-action tests | **PASS** |
| UR2.13 | Receive explanations of proposed actions before execution | Preview/explanation cards before confirm | TE2; preview/confirm tests | **PASS** |
| UR2.14 | Create Canvas assignments | Assignment create form + assistant create_assignment | TE2; assignment tests | **PASS** |
| UR2.15 | Update assignment details and dates | Assignment update + due-date normalization | TE2; assignment update tests | **PASS** |
| UR2.16 | Create Canvas quizzes | Quiz create workflow + validation | TE2; quiz tests | **PASS** |
| UR2.17 | List submissions for assignments | Submissions pages + Canvas submissions API | TE2; submission tests | **PASS** |
| UR2.18 | Draft Canvas messages to selected students | Messages compose/draft + templates | TE2; messaging tests | **PASS** |
| UR2.19 | Review and edit messages before sending | Message editor/preview before confirm | TE2; messaging tests | **PASS** |
| UR2.20 | Draft feedback comments for selected submissions | Feedback create/draft workflow | TE2; feedback tests | **PASS** |
| UR2.21 | Review and approve feedback before posting | Feedback review + confirm gate | TE2; feedback tests | **PASS** |
| UR2.22 | Modify grades through approved workflows | Grade preview/submit + strong CONFIRM | TE2; grade tests | **PASS** |
| UR2.23 | Preview Canvas-modifying actions before execution | Dry-run / write preview services | TE2; preview tests | **PASS** |
| UR2.24 | Explicitly approve Canvas-modifying actions before execution | Confirm controls; no auto-write on chat turn | TE2; confirm/scope tests | **PASS** |
| UR2.25 | Cancel a proposed action before execution | Cancel path leaves Canvas unchanged | TE2; cancel/e2e tests | **PASS** |
| UR2.26 | View a history of previous actions | Workflow/action history UI | TE2; history tests | **PASS** |
| UR2.27 | View previous assistant requests and outcomes | Conversation/history + workflow status | TE2; history/chat tests | **PASS** |
| UR2.28 | Determine whether a previous action can be reversed | Reversal metadata + undo for supported creates | TE2; undo/reversal tests | **PASS** |
| UR2.29 | Clear feedback when requests cannot be completed safely | Clarifications, errors, out-of-scope refusals | TE2; error-handling tests | **PASS** |

## 4. Functional Requirements Traceability

Functional requirements are grouped as in the TE2 matrix. Each item was recorded PASS under unit, integration, system, usability, acceptance, and/or operational testing as applicable.

### *FR1 — Authentication, Accounts & Canvas Credentials*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR1.1 | Instructor login required before Canvas functionality | Auth + route guards | TE2 PASS | **PASS** |
| FR1.2 | Administrators manage instructor accounts | Admin users service/UI | TE2 PASS | **PASS** |
| FR1.3 | Administrators can revoke instructor access | Revoke/disable flows | TE2 PASS | **PASS** |
| FR1.4 | Instructors enter/replace/remove Canvas tokens | Connect Canvas / settings | TE2 PASS | **PASS** |
| FR1.5 | Canvas API tokens encrypted before storage | Fernet encryption | TE2 PASS | **PASS** |
| FR1.6 | Canvas API tokens validated before use | Token validation service | TE2 PASS | **PASS** |
| FR1.7 | Stored tokens never exposed after submission | Masked UI; no token echo | TE2 PASS | **PASS** |
| FR1.8 | Canvas ops use authenticated instructor token | Per-instructor connections | TE2 PASS | **PASS** |

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR1.9 | No cross-instructor token access | Ownership checks | TE2 PASS | **PASS** |
| FR1.10 | Logout and session expiration supported | Sessions + revoke on logout/password change | TE2 PASS | **PASS** |

### *FR2 — Course Dashboard & Canvas Data Access*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR2.1 | Retrieve instructor courses | Canvas course sync/list | TE2 PASS | **PASS** |
| FR2.2 | Select active course | Active-course preference API/UI | TE2 PASS | **PASS** |
| FR2.3 | Retrieve students for selected course | Students service/pages | TE2 PASS | **PASS** |
| FR2.4 | Retrieve assignments for selected course | Assignments service/pages | TE2 PASS | **PASS** |
| FR2.5 | Retrieve submissions for assignments | Submissions service/pages | TE2 PASS | **PASS** |
| FR2.6 | Dashboard course-level summaries | Instructor/course overview | TE2 PASS | **PASS** |
| FR2.7 | Assignment-level info and submission status | Course assignments/overview | TE2 PASS | **PASS** |
| FR2.8 | Identify submitted/missing/late/graded where available | Submission analysis | TE2 PASS | **PASS** |
| FR2.9 | Suggested actions / workflow shortcuts | Dashboard quick actions | TE2 PASS | **PASS** |
| FR2.10 | Canvas is source of truth for course/assignment data | Sync model; no local override of Canvas truth | TE2 PASS | **PASS** |

### *FR3 — Natural-Language Assistant*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR3.1 | Chat-style NL interface | Assistant page + ChatWindow | TE2 PASS | **PASS** |
| FR3.2 | Support approved Canvas course-management workflows | Allowlisted actions | TE2 PASS | **PASS** |
| FR3.3 | Interpret requests into structured actions | Chat/LLM structured output | TE2 PASS | **PASS** |
| FR3.4 | Request clarification when incomplete/ambiguous | Clarification responses | TE2 PASS | **PASS** |
| FR3.5 | Display interpreted intent before execution | Proposal/preview cards | TE2 PASS | **PASS** |
| FR3.6 | Support listed read/write workflows (courses, assignments, messages, quizzes, grades, etc.) | Tool schemas + UI handoffs | TE2 PASS | **PASS** |
| FR3.7 | Explain unsupported requests and prevent execution | out_of_scope / refusal paths | TE2 PASS | **PASS** |
| FR3.8 | Must not invent Canvas data not retrieved from Canvas | Feature 13 grounding / ID scrub | TE2 PASS | **PASS** |
| FR3.9 | Scope to active/explicit course context | Course resolution priority | TE2 PASS | **PASS** |

### *FR4 — Assignment & Quiz Management*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR4.1 | Create Canvas assignments | Assignment create services/UI | TE2 PASS | **PASS** |
| FR4.2 | Update assignment details including dates | Assignment update + due_date | TE2 PASS | **PASS** |
| FR4.3 | Create Canvas quizzes | Quiz create services/UI | TE2 PASS | **PASS** |
| FR4.4 | Validate assignment/quiz data before Canvas submit | Validation services | TE2 PASS | **PASS** |
| FR4.5 | Require preview and confirmation for create/modify | Write preview + confirm | TE2 PASS | **PASS** |

### *FR5 — Submission Analysis & Grading*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR5.1 | Identify missing submissions | Missing analysis service/UI | TE2 PASS | **PASS** |
| FR5.2 | Identify late submissions | Late analysis service/UI | TE2 PASS | **PASS** |
| FR5.3 | Display submission summaries and affected students | Analysis tables/summaries | TE2 PASS | **PASS** |

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR5.4 | Support grade modification via approved operations | Grade management services | TE2 PASS | **PASS** |
| FR5.5 | Grade modifications require preview and confirmation | Grade preview + CONFIRM | TE2 PASS | **PASS** |

### *FR6 — Messaging & Feedback*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR6.1 | Generate draft Canvas messages | Message draft services/UI | TE2 PASS | **PASS** |
| FR6.2 | Review/edit messages before sending | Message editor | TE2 PASS | **PASS** |
| FR6.3 | Message delivery requires preview and confirmation | Send confirm gate | TE2 PASS | **PASS** |
| FR6.4 | Generate feedback comments for submissions | Feedback draft workflow | TE2 PASS | **PASS** |
| FR6.5 | Feedback posting requires preview and confirmation | Feedback confirm gate | TE2 PASS | **PASS** |
| FR6.6 | Messaging and feedback recorded in action history | Workflow/history logging | TE2 PASS | **PASS** |

### *FR7 — Dry-Run Preview & Confirmation*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR7.1 | Every Canvas-modifying action generates dry-run preview | write_preview / confirm flow | TE2 PASS | **PASS** |
| FR7.2 | Preview summarizes action and affected resources | Preview panel cards | TE2 PASS | **PASS** |
| FR7.3 | Preview identifies reversibility | Reversibility status | TE2 PASS | **PASS** |
| FR7.4 | Explicit confirm required before Canvas write | Confirm controls | TE2 PASS | **PASS** |
| FR7.5 | Users can cancel proposed actions | Cancel path | TE2 PASS | **PASS** |
| FR7.6 | Classify actions as read-only, modifying, or irreversible | Action classification | TE2 PASS | **PASS** |

### *FR8 — Action History & Limited Reversal*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR8.1 | Maintain user-visible action history | History pages/API | TE2 PASS | **PASS** |
| FR8.2 | History includes request, action, status, timestamp, affected resources | Workflow history model | TE2 PASS | **PASS** |
| FR8.3 | Maintain history of assistant requests and associated actions | Conversation/workflow linkage | TE2 PASS | **PASS** |
| FR8.4 | Support limited reversal where safe/practical | Undo for supported creates | TE2 PASS | **PASS** |
| FR8.5 | Explain when reversal is unavailable | Reversal metadata messaging | TE2 PASS | **PASS** |

### *FR9 — AI Validation & MCP Integration*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR9.1 | AI model must not directly access Canvas APIs | LLM → backend tools only | TE2 PASS | **PASS** |
| FR9.2 | Canvas operations routed through backend and MCP/tool layer | Dispatcher + optional MCP | TE2 PASS | **PASS** |
| FR9.3 | Backend validates AI-generated actions before execution | Validation before confirm/execute | TE2 PASS | **PASS** |
| FR9.4 | Reject malformed/unsupported/unauthorized/unsafe actions | Allowlists + validators | TE2 PASS | **PASS** |
| FR9.5 | MCP/tool layer exposes only approved Canvas operations | MCP allowlist/host controls | TE2 PASS | **PASS** |
| FR9.6 | Dockerized MCP server or equivalent tool-execution layer | app/mcp + compose | TE2 PASS | **PASS** |

### *FR10 — Security & Privacy*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR10.1 | Encrypt Canvas tokens and AI API keys at rest | Fernet / encrypted columns | TE2 PASS | **PASS** |
| FR10.2 | Credentials never in logs, errors, responses, repos, or UI | Redaction + masking | TE2 PASS | **PASS** |
| FR10.3 | Student information only for authorized users | AuthZ / course ownership | TE2 PASS | **PASS** |
| FR10.4 | Minimize student-identifiable info sent to AI | Student context minimization | TE2 PASS | **PASS** |
| FR10.5 | Validate all Canvas-modifying actions before forwarding | Backend validation gate | TE2 PASS | **PASS** |

### *FR11 — Reliability & Error Handling*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR11.1 | Meaningful user-facing error messages | Error classification + UI alerts | TE2 PASS | **PASS** |
| FR11.2 | Safely handle Canvas API failures | Failure recording; no fake success | TE2 PASS | **PASS** |
| FR11.3 | Safely handle incomplete/unavailable Canvas data | Graceful empty/partial states | TE2 PASS | **PASS** |
| FR11.4 | Prevent duplicate Canvas-modifying execution | Idempotency keys / write guards | TE2 PASS | **PASS** |

### *FR12 — Administration & Deployment*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR12.1 | Administrator operational health information | System health page | TE2 PASS | **PASS** |
| FR12.2 | Monitor failed Canvas and AI operations | Failed operations page | TE2 PASS | **PASS** |
| FR12.3 | Web-based application | React + FastAPI + nginx compose | TE2 PASS | **PASS** |
| FR12.4 | Docker Compose for local deployment | docker-compose.yml / Makefile | TE2 PASS | **PASS** |
| FR12.5 | Developer setup and deployment documentation | docs/final deployment + developer guides | TE2 PASS | **PASS** |
| FR12.6 | Separated frontend, backend, DB, Canvas, MCP components | Repository architecture | TE2 PASS | **PASS** |

### *FR13 — Scope Constraints*

| ID | Requirement | Implementation evidence | Validation evidence | Status |
|---|---|---|---|---|
| FR13.1 | No autonomous Canvas modifications without approval | Confirm-before-write; no write on normal chat turn | TE2 PASS | **PASS** |
| FR13.2 | No Canvas API execution bypassing backend validation | Allowlisted tools only | TE2 PASS | **PASS** |
| FR13.3 | No institution-wide Canvas administration | Instructor-scoped token/course | TE2 PASS | **PASS** |
| FR13.4 | No unrestricted deletion of Canvas resources | Deletes limited to undo paths | TE2 PASS | **PASS** |
| FR13.5 | No arbitrary user-defined Canvas API calls | No free-form Canvas proxy | TE2 PASS | **PASS** |

## 5. Milestone Coverage (Implementation Path)

Requirements were delivered and validated incrementally across project milestones. The following summarizes where major requirement groups reached implementation and validation readiness:

| Milestone | Focus | Requirements advanced |
|---|---|---|
| M1 — Proof of Life (Jun 12) | Foundation: FE/BE, auth, Canvas, local AI, Docker | Foundation for UR2.1–2.3, 2.11–2.12; FR1–FR3, FR12–FR13 |
| M2 — MVP Demo (Jul 7) | Auth, Canvas data, assistant, assignments, messaging, preview, history | UR1.1–1.3; UR2.1–2.15, 2.18–2.19, 2.23–2.29; FR1–FR5, FR7–FR8, FR11 |
| M3 — Testing Event 1 (Jul 14) | MVP validation, coverage, usability fixes | Validation of MVP URs/FRs; TE1 evidence |
| M4 — Testing Event 2 (Jul 28) | Quizzes, feedback, grades, reversal + TE2 validation | UR2.16, 2.20–2.22, 2.28; FR4, FR6, FR8; all TE2 PASS |

| Milestone | Focus | Requirements advanced |
|---|---|---|
| M5 — Final Demo (Aug 7) | Admin monitoring, LLM providers, security, reliability, MCP | UR1.4–1.7; FR9–FR12 completion |
| M6 — Final Delivery (Aug 11) | Docs, deployment package, RTM, testing report, submission | FR12.5–12.6 documentation; full UR/FR documentation |

## 6. Evidence Sources

• Testing Event 2 requirements matrix — every listed UR/FR/scope constraint recorded as PASS.
• Final Testing Report — automated suite results (Pytest, Vitest, Playwright, Hypothesis) and feature-level PASS summary.
• Repository automated tests — `make test`, `make test-e2e`, `make test-fuzz`.
• Final documentation package — user guides, developer docs, architecture/API, deployment guides under `docs/final/`.
• Milestone planning/traceability workbook — feature-to-requirement mapping across Milestones 1–6.

## 7. Conclusion

All administrator requirements (UR1.1–UR1.7), instructor requirements (UR2.1–UR2.29), functional requirement groups (FR1–FR13), and scope constraints were implemented in the delivered system and validated as PASS in Testing Event 2. No requirement-level FAIL results remain at final submission. Residual product limitations are intentional scope constraints (FR13), not unmet requirements.